NETWORK DESIGN PROJECT

Enterprise Multi-Branch Network Implementation

Prepared by

Hazem Hamdi Mahmoud Abdelqader

Nahda University (NUB)

Faculty of Engineering | Communications & Electronics Engineering

Contact Information

+20 106 238 6571

hazemhamdypersonal@gmail.com

github.com/Hazem-Hamd

1. Project Overview

This report documents the complete design and implementation of a multi-branch enterprise network. The network serves three geographically separated branches, each hosting five departments, connected through a central ISP infrastructure. The design emphasizes scalability, security, redundancy, and efficient IP address utilization.

The network topology includes:

- Three branch offices, each with Marketing, Accounting, Purchasing, HR, and IT departments
- A centralized Server Farm hosting DHCP, TFTP, and other shared services
- Three border routers (RB1, RB2, RB3) interconnected over WAN links
- VLAN segmentation for traffic isolation between departments
- EtherChannel for aggregated bandwidth and link redundancy
- Secure remote management via SSH and access control via ACLs

2. IP Addressing Strategy — VLSM

The network uses Variable Length Subnet Masking (VLSM) to efficiently divide the base Class C address space. Starting from the parent network 192.168.0.0/24, subnets are allocated by fitting the smallest possible block around each department's device count, eliminating the wasted addresses that Fixed Length Subnet Masking (FLSM) would produce.

VLSM Calculation Method

For each department, the required subnet size is calculated as follows:

- Determine the number of devices (hosts) needed.
- Compute $n = \lceil \log_2(\text{devices}) \rceil$ — the number of host bits required.
- The block size is 2^n, and the subnet mask prefix is /(32 − n).
- Verify the non-blocking rule: (block start address) ÷ (block size) must be an integer.

Example — Marketing (50 hosts):

- $\log_2(50) \approx 5.64 \rightarrow$ rounded up to 6 bits → block size = 2^6 = 64 addresses
- Subnet mask: /26 (32 − 6 = 26 ones in the mask)
- Non-blocking check: 0 ÷ 64 = 0 ✓

Branch 1 — VLAN Addressing (192.168.0.0/24, first half)

VLAN	Department	Devices	Network IP	Subnet Mask	IP Range
3	Marketing	50	192.168.0.0	/26	0 – 63
4	Accounting	25	192.168.0.64	/27	64 – 95
5	Purchasing	13	192.168.0.96	/28	96 – 111

VLAN	Department	Devices	Network IP	Subnet Mask	IP Range
6	HR	2	192.168.0.112	/29	112 – 119
7	IT	1	192.168.0.120	/30	120 – 123
–	Mult-Rout	2	192.168.0.124	/30	124 – 127

Branch 2 — VLAN Addressing (192.168.0.0/24, second half)

VLAN	Department	Devices	Network IP	Subnet Mask	IP Range
8	Marketing	50	192.168.0.128	/26	129 – 191
9	Accounting	25	192.168.0.192	/27	193 – 223
10	Purchasing	13	192.168.0.224	/28	225 – 239
11	HR	2	192.168.0.240	/29	241 – 247
12	IT	1	192.168.0.248	/30	249 – 251
–	Mult-Rout	2	192.168.0.252	/30	253 – 254

Branch 3, WAN Links & Server Farm (192.168.1.0/24)

VLAN	Department	Devices	Network IP	Subnet Mask	IP Range
13	Marketing	50	192.168.1.0	/26	0 – 63
14	Accounting	25	192.168.1.64	/27	64 – 95
15	Purchasing	13	192.168.1.96	/28	96 – 111
16	HR	2	192.168.1.112	/29	112 – 119
17	IT	1	192.168.1.120	/30	120 – 123
–	Mult-Rout	2	192.168.1.124	/30	124 – 127
–	Servers Farm	3	192.168.1.128	/29	128 – 135
–	RB1-ISP	2	192.168.1.136	/30	136 – 139
–	WAN (RB3-RB2)	2	192.168.1.140	/30	141 – 142
–	WAN (RB2-RB1)	2	192.168.1.144	/30	145 – 146
–	RB2-ISP	2	192.168.1.148	/30	149 – 151
–	RB3-ISP	2	192.168.1.152	/30	153 – 155

3. DHCP Implementation

A centralized DHCP server (located at 192.168.1.130 in the Server Farm) dynamically assigns IP addresses to all end-devices. This eliminates manual configuration, prevents address conflicts, and simplifies device mobility between VLANs.

Key Benefits

- **Automation:** The DHCP server assigns IPs from a pre-defined pool the moment a device joins any subnet.
- **Mobility:** Devices moving between VLANs automatically obtain the correct IP address for their new segment without manual intervention.
- **Conflict Prevention:** The server tracks all leased addresses, ensuring uniqueness across the entire network.
- **Centralized Management:** A single server manages all three branches, reducing administrative overhead.

Configuration

Router and SVI (Switch Virtual Interface) gateway addresses are statically assigned. The ip helper-address command on each SVI forwards DHCP broadcast requests to the central server as unicast packets:

```
Enable
conf t
! --- Configure VLAN Gateway Interface ---
interface vlan <VLAN_ID>
! Assign the gateway IP for this VLAN
ip address <GATEWAY_IP> <SUBNET_MASK>
! Forward DHCP broadcasts to the central server
ip helper-address 192.168.1.130
exit
```

4. EtherChannel Implementation

EtherChannel (IEEE 802.3ad / LACP) bundles multiple physical links between switches into a single logical interface, providing two critical advantages for the enterprise network:

- **Increased Bandwidth:** Multiple physical links are aggregated. For example, two 100 Mbps links create an effective 200 Mbps logical channel.
- **Link Redundancy:** If one physical cable fails, traffic seamlessly continues over the remaining links with no network interruption.

LACP Active mode is used on all channel groups to ensure dynamic negotiation and automatic failover. The logical port-channel interface is configured as a trunk to carry all VLAN traffic between switches:

```
Enable
conf t
! Step 1: Select the physical ports to bundle
```

```
interface range fastEthernet 0/1-2
! Step 2: Assign to channel group using LACP Active mode
channel-group 1 mode active
no shutdown
exit
! Step 3: Configure the logical port-channel interface
interface port-channel 1
! Configure as trunk to carry all VLANs
switchport mode trunk
switchport trunk encapsulation dot1q
no shutdown
exit
```

5. Port Security

Port Security is configured on all access ports to prevent unauthorized devices from connecting to the network. The switch memorizes the MAC address of the first authorized device on each port. If a device with a different MAC address attempts to connect, the port is immediately shut down (Shutdown violation mode).

Static MAC address assignment is used for known workstations to provide deterministic control:

```
Enable
conf t
! Step 1: Select the access port
interface fastEthernet 0/1
! Step 2: Set to access mode
switchport mode access
! Step 3: Enable port security
switchport port-security
! Step 4: Assign the authorized MAC address (replace with actual MAC)
switchport port-security mac-address 00E0.F725.337A
! Step 5: Shutdown the port on any violation
switchport port-security violation shutdown
exit
! Verification
show port-security interface fastEthernet 0/1
```

6. VTP — VLAN Trunking Protocol

In an enterprise with 20+ switches, manually creating VLANs on each device is error-prone and time-consuming. VTP (VLAN Trunking Protocol) solves this by using a server-client model:

- One central switch acts as the VTP Server. VLANs are created only here.
- All other switches operate as VTP Clients, automatically downloading the VLAN database from the server over trunk links.
- A shared domain name and optional password prevent unauthorized switches from joining the VTP domain.

VTP Server Configuration (Main Switch)

```
Enable
conf t
! Set the VTP domain name (must match on all switches)
vtp domain amazon
! Set a password for security
vtp password 123
! Set this switch as the VTP Server
vtp mode server
! Create VLANs once here — they propagate automatically
vlan 10
name IT
vlan 20
name HR
vlan 30
name Sales
exit
```

VTP Client Configuration (Access Switches)

```
Enable
conf t
! Match the domain name exactly
vtp domain amazon
! Match the password
vtp password 123
! Set to client mode
vtp mode client
! Ensure the uplink to the server is a trunk
interface fastEthernet 0/1
switchport mode trunk
exit
! Verification commands
show vtp status
show vlan brief
```

7. Secure Remote Access — SSH

All three border routers (RB1, RB2, RB3) are configured for SSH-only remote management. SSH encrypts the entire session — credentials, commands, and responses — making it impossible for network eavesdroppers to capture sensitive configuration data.

Access is further restricted by an ACL that permits only the IT administrator's workstation (192.168.1.122) to initiate SSH sessions. All other source IPs are denied at the VTY lines:

```
Enable
conf t
! Step 1: Set hostname and domain (required for RSA key generation)
hostname RB1
ip domain-name amazon.com
! Step 2: Generate 1024-bit RSA encryption keys
```

```
crypto key generate rsa
! Step 3: Create the admin account with full privilege level 15
username IT1 privilege 15 secret 123
! Step 4: Create an ACL — permit only the IT admin workstation
access-list 1 permit host 192.168.1.122
! Step 5: Apply to VTY lines — SSH only, no Telnet
line vty 0 4
login local
transport input ssh
access-class 1 in
exit
! To connect: ssh -l IT1 192.168.X.X
```

8. Routing — RIP Version 2

RIP Version 2 (RIPv2) is used as the dynamic routing protocol across all three border routers. Version 2 is specifically required because it supports VLSM (classless routing), allowing the routers to correctly exchange subnet information for all the variable-length subnets defined in the IP addressing scheme.

Why RIPv2 over RIPv1

- **VLSM Support:** RIPv1 is classful and would collapse /26, /27, /28, /29, and /30 subnets back to /24, causing routing failures. RIPv2 includes the subnet mask in its updates.
- **No Auto-Summary:** The no auto-summary command is critical. Without it, RIPv2 would automatically summarize routes to their classful boundary, defeating the purpose of VLSM.

```
Enable
conf t
! Enter RIP configuration mode
router rip
! Enable Version 2 for VLSM support
version 2
! CRITICAL: Disable auto-summary to preserve VLSM subnet masks
no auto-summary
! Advertise all directly connected networks
network 192.168.0.0
network 192.168.1.0
exit
```

9. TFTP — Configuration Backup & Restore

The TFTP (Trivial File Transfer Protocol) server in the Server Farm (192.168.1.130) acts as a centralized configuration backup system. Each router's running-config is uploaded to the TFTP server with a unique, timestamped filename. In the event of hardware failure, a replacement router can be fully restored in minutes.

Backup — Upload Configuration to TFTP Server

```
Enable
! Optional: verify connectivity to the TFTP server
ping 192.168.1.130
! Copy the running configuration to the TFTP server
copy running-config tftp:
! Answer the interactive prompts:
! Address or name of remote host? -> 192.168.1.130
! Destination filename? -> RB1-Backup-Jan2026
```

Restore — Download Configuration from TFTP Server

```
Enable
! Copy the backup FROM the TFTP server TO the running config
copy tftp: running-config
! Answer the interactive prompts:
! Address or name of remote host? -> 192.168.1.130
! Source filename? -> RB1-Backup-Jan2026
! Destination filename? -> [Press ENTER to accept default]
```

10. Default Route to ISP

A default static route on each border router forwards any packet with an unknown destination to the ISP, providing internet connectivity. The route is then redistributed into RIP using the default-information originate command, so that all branch routers learn the path to the internet automatically:

```
Enable
conf t
! 0.0.0.0 0.0.0.0 matches ANY destination IP / ANY subnet mask
! Serial0/1/0 is the exit interface toward the ISP
ip route 0.0.0.0 0.0.0.0 Serial0/1/0
! Redistribute the default route into RIP
router rip
! This tells neighboring routers: 'I am the gateway to the internet'
default-information originate
exit
```

11. Access Control — Restricting ICMP (Ping)

By default, any device can ping any other device, which reveals network topology to potential attackers and generates unnecessary traffic. An Extended ACL is applied to restrict ICMP (ping) traffic to the IT administrator only, while allowing all other legitimate traffic to flow unimpeded.

ACL Logic

- **Rule 1 — Allow IT Admin pings:** Permit ICMP from 192.168.1.122 to any destination.

- **Rule 2 — Allow echo-reply:** Permit ICMP echo-reply packets so ping responses return to authorized senders (without this, pings from the IT admin would never receive answers).
- **Rule 3 — Allow unreachable:** Permit ICMP unreachable messages for network diagnostics.
- **Rule 4 — Block all other pings:** Deny all remaining ICMP requests from any other source.
- **Rule 5 — Permit all other IP traffic:** CRITICAL. Without this implicit permit, the ACL would block all internet and application traffic for every user.

```
Enable
conf t
! Rule 1: Allow IT Admin to ping anyone
access-list 101 permit icmp host 192.168.1.122 any
! Rule 2: Allow echo-reply so IT Admin receives ping responses
access-list 101 permit icmp any any echo-reply
! Rule 3: Allow ICMP unreachable for diagnostics
access-list 101 permit icmp any any unreachable
! Rule 4: Block all other ICMP (ping) requests
access-list 101 deny icmp any any
! Rule 5: CRITICAL — Allow all other IP traffic (web, email, etc.)
access-list 101 permit ip any any
! Apply the ACL inbound on each VLAN interface
interface vlan <VLAN_ID>
ip access-group 101 in
exit
```

12. Summary

This network design delivers a robust, scalable, and secure multi-branch enterprise infrastructure. The table below summarizes the key technologies deployed:

Technology	Purpose & Benefit
VLSM	Efficient IP address allocation across 3 branches, 5 departments each — eliminates wasted address space
VLAN	Layer 2 traffic isolation between departments for security and broadcast control
DHCP	Automatic IP assignment from a central server, preventing conflicts and simplifying administration
EtherChannel	Aggregated link bandwidth and automatic failover between switches
Port Security	Prevents unauthorized devices from connecting via MAC address enforcement
VTP	Centralized VLAN database management propagated automatically to all client switches
SSH	Encrypted remote management with ACL-restricted access to IT admin only

Technology	Purpose & Benefit
RIPv2	Dynamic routing with VLSM support across all branch routers
TFTP Backup	Centralized router configuration backup and rapid restore capability
Default Route	Internet gateway advertised to all branches via RIP redistribution
Extended ACL	Fine-grained ICMP access control restricting ping to authorized administrators

End of Report

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